

Flow, Pressure & Resistance

The hydraulic–electrical analogy — a Semitree cheat sheet

For laminar flow, a chip behaves like a circuit: pressure drives flow through hydraulic resistance. Pressure ↔ voltage, flow rate ↔ current, resistance ↔ resistance.

Ohm's law for fluids

$$\Delta P = Q \cdot R$$

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the master relation between pressure, flow rate, and resistance

Resistance (circular)

$$R$$

$$R = 128 \cdot \mu \cdot L / (\pi \cdot D^4)$$

exact for fully developed laminar flow

Resistance (rectangular)

$$R \approx$$

$$R \approx 12 \cdot \mu \cdot L / (w \cdot h^3 \cdot (1 - 0.63 \cdot h/w))$$

APPROXIMATION — most accurate for $h \ll w$

Series channels

$$\Sigma$$

$$R = R_1 + R_2 + \dots$$

resistances add end-to-end

Parallel channels

$$\square$$

$$1/R = 1/R_1 + 1/R_2 + \dots$$

reciprocals add

Hydraulic diameter

$$D_h$$

$$D_h = 2 \cdot w \cdot h / (w + h)$$

rectangular cross-section

Variables

ΔP pressure drop (Pa)

Q flow rate (m³/s)

R resistance (Pa·s/m³)

μ dynamic viscosity (Pa·s)

L channel length (m)

D diameter (m)

w width (m)

h height (m)

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Standard, published equations. Always verify against primary sources for critical work.